23.7). Cooper then travels that 1 AU distance through the bulk, parallel to our brane, to reach the Earth and visit Murph; see Figure 29.4.

DANGER: The Sandwich Is Unstable



In 2006, I used Einstein's relativistic laws to work out a mathematical description of the AdS layer and its confining branes. Because I had never before worked with relativity in five dimensions, I asked Lisa Randall to critique my analysis. Lisa browsed it quickly, and then told me some good news and some bad news.

The good news: My idea of an AdS sandwich had been invented six years earlier by Ruth Gregory (University of Durham, UK), together with Valery Rubakov and Sergei Sibiryakov (Institute for Nuclear Research in Moscow, Russia). This showed I was not being stupid in my first mathematical foray into the bulk. I had rediscovered something worth discovering.

The bad news: Edward Witten (Princeton) and others had shown that the AdS sandwich is *unstable!* The confining branes are under pressure, rather like a playing card that you squeeze end to end between your finger and thumb (Figure 23.8). The card bends, and with further squeezing, it buckles. Similarly, the confining branes will bend and crash into our brane (our universe), destroying it. The entire universe destroyed! That's the worst news ever!!

But I can think of several ways to save our universe, if it really does live in an AdS sandwich (which I very much doubt it does); several ways to "stabilize the confining branes," in the jargon of physicists.

In my science interpretation of *Interstellar*, Professor Brand, working with Einstein's relativity equations, rediscovers the AdS sandwich, as I did; see the photograph of his blackboard in Figure 3.6.